

Leo Chen

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EDUCATION

Santa Clara University

Santa Clara, CA

Master of Science & Bachelor of Science in Electrical and Computer Engineering

Sept 2022 – Jun 2026

Activities: Institute of Electrical and Electronics Engineers (IEEE), Association of Computing Machinery (ACM)

Courses: Data Structures, Algorithms, Operating Systems, Artificial Intelligence, Computer Architecture, Logic Design, Embedded C & Assembly, Probability & Statistics, Microelectronic Circuits, Circuits II

SKILLS

Languages: Python, C/C++, Verilog, JavaScript/TypeScript, HTML/CSS, Matlab, Shell Script

Frameworks: PyTorch, Tensorflow, Keras, Numpy, Matplotlib, Angular

Tools: Synopsys VCS & Design Compiler, Linux, Git, Docker, Google Cloud, Firebase, VS Code, JetBrains Suite

Interests: Hiking, Running, Lifting, Snowboarding, Photography, Traveling, Cars

EXPERIENCE

Undergraduate Machine Learning Researcher

Jun 2024 – Present

Santa Clara University

Santa Clara, CA

- Developed modular vision transformer models using TensorFlow, Keras, and PyTorch on Linux systems.
- Researching Chain-of-Thought prompting and Backbone Branching to enhance detection transformers.
- Optimized detection transformers for satellite imagery via layer modifications and hyperparameter tuning.
- Benchmarking models through ablation studies to improve inference speed and detection accuracy.

Executive Board Member

Mar 2024 – Present

IEEE at Santa Clara University

Santa Clara, CA

- Collaborated with professors and industry professionals to provide 200+ students with career & academic insights
- Designed quarterly university wide software & firmware workshops to spark interest in electrical engineering
- Boosted club participation by through weekly newsletter and building webpages reaching 400+ subscribers

Mentor

Aug 2023 – Jul 2024

Santa Clara University

Santa Clara, CA

- Provided structured technical guidance to incoming students, focusing on software engineering best practices, project design, and debugging strategies.
- Led one on one mentoring sessions on academic and professional guidance

PROJECTS

Advanced Logic Design at Santa Clara University | Verilog, Synopsys VCS, Design Compiler

Mar 2024

- Implemented complex digital state machines in Verilog, including a game controller utilizing state transitions and bitwise encoding to ensure correct functionality.
- Designed and verified an SRAM-like SPI interface, replicating serial communication and data transmission using precise timing constraints and thorough testbench validation.
- Developed and validated a sine-wave generator, employing interpolative phase accumulators, ROM lookup tables, and precision numeric representations; confirmed accuracy through testbench and C-model comparisons.

Fist Bump | 3rd Place Hack for Humanity 2024

Feb 2024

- Developed a web platform to streamline student event tracking with a calendar view using Dart's Shelf library
- Engineered a unique backend database solution using Dart's Shelf library with 5 different REST API endpoints

The Marina Project | 3rd Place ACM Winter Challenge

Dec 2023

- Engineered a full-stack mobile application using Flutter and Dart
- Integrated 12+ REST APIs to curate music based on geographic location
- Optimized app responsiveness with data caching for a smooth user experience, improved response time by 30%
- Implemented future loading structures to facilitate smooth transition between pages